

Partial Progress on Nonzero Residuals in a Relative Stopping Criterion for the Inexact Bregman Proximal–Gradient Method

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Abstract

Yang and Toh’s partial relative stopping criterion for the inexact Bregman proximal–gradient method assumes that the primal-dual residual in the approximate subproblem optimality condition is zero. This note gives a rigorous way to permit nonzero residuals in the non-inertial method. The main observation is that the residual term in the basic descent estimate need not be bounded in absolute value: it can be kept with its sign and controlled uniformly over a comparison set by a support-gap quantity. The resulting support-gap relative stopping criterion preserves the same telescoping descent mechanism as the zero-residual criterion and yields objective convergence, an ergodic $O(1/k)$ bound, a last-shadow-iterate $O(1/k)$ bound under the usual additional condition, and whole-sequence convergence under standard Bregman consistency assumptions.

The result should be read as partial progress, not as a full solution of the broad future-research question in Yang–Toh. It treats the non-inertial iBPGM and allows nonzero residuals only through a comparison-set support-gap bound. On an unbounded full comparison set this bound is finite only in special residual directions, and for $C = \mathcal{E}$ it forces the residual to be zero. The inertial method is not addressed here.

1 Status of the result

The result below is directly related to the obstruction in Yang and Toh’s analysis of their partial relative stopping criterion: in the descent inequality, the inexact subproblem residual produces a term of the form

$$\langle \Delta^k, \tilde{x}^{k+1} - x \rangle.$$

Their partial relative criterion removes this term by imposing $\Delta^k = 0$. The present note removes that restriction in a conditional way. It introduces a comparison set C and requires that the positive part of the residual, uniformly over $x \in C$, be small relative to the Bregman step. This is mathematically meaningful, because it gives the same clean descent inequality used in the zero-residual proof.

It is not a complete resolution of the most ambitious interpretation of the open problem. First, all guarantees are with respect to C ; they are global only if C contains a global minimizer and the shadow iterates remain in C . Second, if $C = \mathcal{E}$, the support-gap condition permits only $\Delta^k = 0$. Third, no inertial analogue is proved. Thus the contribution is best described as a rigorous localized extension of the non-inertial relative-error theory.

2 Setting and notation

Let $\mathcal{E}$ be a finite-dimensional Euclidean space with inner product $\langle \cdot, \cdot \rangle$. We consider the convex composite problem

$$\min_{x \in \mathcal{E}} F(x) := P(x) + f(x), \tag{1}$$

where $P : \mathcal{E} \rightarrow (-\infty, +\infty]$ is proper, closed, and convex, and $f : \mathcal{E} \rightarrow (-\infty, +\infty]$ is proper, closed, and convex, differentiable on the points where its gradient is used. Feasibility constraints may be absorbed into P through an indicator function.

Let $\phi : \mathcal{E} \rightarrow (-\infty, +\infty]$ be a proper closed convex kernel whose interior domain

$$U := \text{int dom } \phi$$

is nonempty and on which ϕ is differentiable. The associated Bregman distance is

$$D_\phi(u, v) := \phi(u) - \phi(v) - \langle \nabla \phi(v), u - v \rangle, \quad u \in \text{dom } \phi, v \in U. \quad (2)$$

We assume that f is L -smooth relative to ϕ on the region visited by the method, meaning that

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + LD_\phi(y, x) \quad (3)$$

whenever $x \in U$ and y is one of the shadow points appearing below. In particular, the standard global relative-smoothness assumption implies (3).

Fix a stepsize parameter

$$\lambda > L. \quad (4)$$

The exact Bregman proximal-gradient subproblem at $x^k \in U$ is

$$\min_{x \in \mathcal{E}} P(x) + \langle \nabla f(x^k), x - x^k \rangle + \lambda D_\phi(x, x^k). \quad (5)$$

The inexact framework uses two points, a computed point $x^{k+1} \in U$ and a shadow point $\tilde{x}^{k+1} \in \text{dom } P \cap \text{dom } f \cap \text{dom } \phi$, together with an error pair $(\Delta^k, \delta_k) \in \mathcal{E} \times \mathbb{R}_+$ satisfying

$$\Delta^k \in \partial_{\delta_k} P(\tilde{x}^{k+1}) + \nabla f(x^k) + \lambda(\nabla \phi(x^{k+1}) - \nabla \phi(x^k)), \quad (6)$$

where

$$\partial_\varepsilon P(z) := \{p \in \mathcal{E} : P(u) \geq P(z) + \langle p, u - z \rangle - \varepsilon \text{ for all } u \in \mathcal{E}\}.$$

This is the same two-point approximate optimality relation used in the iBPGM analysis of Yang–Toh.

3 The support-gap relative stopping criterion

Let $C \subset \text{dom } P \cap \text{dom } f \cap \text{dom } \phi$ be a nonempty closed convex comparison set. Throughout the analysis we assume

$$\tilde{x}^k \in C \quad \text{for every } k \geq 1. \quad (7)$$

When last-iterate estimates are stated, we also use the harmless convention $\tilde{x}^0 = x^0$ and assume $x^0 \in C$.

For $z \in C$ and $r \in \mathcal{E}$, define the support gap

$$\omega_C(r; z) := \sup_{u \in C} \langle r, z - u \rangle = \langle r, z \rangle - \inf_{u \in C} \langle r, u \rangle. \quad (8)$$

Since $z \in C$, one has $\omega_C(r; z) \geq 0$, with the value $+\infty$ allowed.

Definition 3.1 (support-gap relative stopping criterion). *Let σ satisfy*

$$0 \leq \sigma < \frac{\lambda - L}{\lambda}. \quad (9)$$

We say that the approximate subproblem solution satisfies SG-ReSC_C if

$$\lambda^{-1} \delta_k + D_\phi(\tilde{x}^{k+1}, x^{k+1}) + \lambda^{-1} \omega_C(\Delta^k; \tilde{x}^{k+1}) \leq \sigma D_\phi(\tilde{x}^{k+1}, x^k). \quad (10)$$

This criterion reduces to the original zero-residual relative criterion when $\Delta^k = 0$. Indeed, then $\omega_C(0; \tilde{x}^{k+1}) = 0$, and (10) controls only δ_k and $D_\phi(\tilde{x}^{k+1}, x^{k+1})$.

Remark 3.2 (norm-sufficient version). *If C has diameter at most R_C , then*

$$\omega_C(r; z) \leq R_C \|r\| \quad (r \in \mathcal{E}, z \in C). \quad (11)$$

Hence the more elementary condition

$$\lambda^{-1} \delta_k + D_\phi(\tilde{x}^{k+1}, x^{k+1}) + \frac{R_C}{\lambda} \|\Delta^k\| \leq \sigma D_\phi(\tilde{x}^{k+1}, x^k) \quad (12)$$

implies SG-ReSC_C. This is often the simplest implementable sufficient condition in bounded feasible or localized settings.

Remark 3.3 (why unbounded sets are different). *The support gap is exactly the quantity needed to control the signed residual uniformly over C :*

$$\sup_{x \in C} \langle \Delta^k, \tilde{x}^{k+1} - x \rangle = \omega_C(\Delta^k; \tilde{x}^{k+1}).$$

If $C = \mathcal{E}$ and $\Delta^k \neq 0$, then taking $u = \tilde{x}^{k+1} - t\Delta^k$ gives

$$\langle \Delta^k, \tilde{x}^{k+1} - u \rangle = t \|\Delta^k\|^2 \rightarrow +\infty.$$

Thus $\omega_{\mathcal{E}}(\Delta^k; \tilde{x}^{k+1}) < \infty$ forces $\Delta^k = 0$. This explains why the result is localized: a nonzero residual can be admitted only when the comparison set gives a finite support gap in the residual direction, or when a different summability-type argument is used.

4 Signed descent

The next lemma is the main technical point. The residual term is kept with its sign instead of being estimated by an absolute value.

Lemma 4.1 (signed inexact descent). *Assume (3). Suppose that $x^k, x^{k+1} \in U$, that $\tilde{x}^{k+1} \in \text{dom } P \cap \text{dom } f \cap \text{dom } \phi$, and that (6) holds with $\delta_k \geq 0$. Then, for every $x \in \text{dom } P \cap \text{dom } f \cap \text{dom } \phi$,*

$$\begin{aligned} F(\tilde{x}^{k+1}) - F(x) &\leq \lambda D_\phi(x, x^k) - \lambda D_\phi(x, x^{k+1}) - (\lambda - L) D_\phi(\tilde{x}^{k+1}, x^k) \\ &\quad + \lambda D_\phi(\tilde{x}^{k+1}, x^{k+1}) + \delta_k + \langle \Delta^k, \tilde{x}^{k+1} - x \rangle. \end{aligned} \quad (13)$$

Consequently, if $\tilde{x}^{k+1} \in C$ and SG-ReSC_C holds, then for every $x \in C$,

$$F(\tilde{x}^{k+1}) - F(x) \leq \lambda D_\phi(x, x^k) - \lambda D_\phi(x, x^{k+1}) - \mu D_\phi(\tilde{x}^{k+1}, x^k), \quad (14)$$

where

$$\mu := \lambda - L - \lambda\sigma > 0. \quad (15)$$

Proof. Choose

$$p^k \in \partial_{\delta_k} P(\tilde{x}^{k+1})$$

such that

$$\Delta^k = p^k + \nabla f(x^k) + \lambda(\nabla \phi(x^{k+1}) - \nabla \phi(x^k)).$$

The definition of $\partial_{\delta_k} P$ gives

$$P(\tilde{x}^{k+1}) - P(x) \leq \langle p^k, \tilde{x}^{k+1} - x \rangle + \delta_k. \quad (16)$$

By convexity of f at x^k and relative smoothness from x^k to $\tilde{x}^{k+1}$,

$$f(\tilde{x}^{k+1}) - f(x) \leq \langle \nabla f(x^k), \tilde{x}^{k+1} - x \rangle + LD_\phi(\tilde{x}^{k+1}, x^k). \quad (17)$$

Adding (16) and (17), and then substituting the expression for p^k , yields

$$\begin{aligned} F(\tilde{x}^{k+1}) - F(x) &\leq \langle \Delta^k, \tilde{x}^{k+1} - x \rangle - \lambda \langle \nabla \phi(x^{k+1}) - \nabla \phi(x^k), \tilde{x}^{k+1} - x \rangle \\ &\quad + LD_\phi(\tilde{x}^{k+1}, x^k) + \delta_k. \end{aligned}$$

The three-point Bregman identity gives

$$\begin{aligned} -\langle \nabla \phi(x^{k+1}) - \nabla \phi(x^k), \tilde{x}^{k+1} - x \rangle &= D_\phi(x, x^k) - D_\phi(x, x^{k+1}) \\ &\quad - D_\phi(\tilde{x}^{k+1}, x^k) + D_\phi(\tilde{x}^{k+1}, x^{k+1}). \end{aligned}$$

Substitution proves (13). If $x \in C$, then

$$\langle \Delta^k, \tilde{x}^{k+1} - x \rangle \leq \omega_C(\Delta^k; \tilde{x}^{k+1}).$$

Combining this bound with SG-ReSC_C gives (14). $\square$

5 Objective convergence and rates

Define the comparison-set value and solution set by

$$F_C^* := \inf_{x \in C} F(x), \quad S_C := \arg \min_{x \in C} F(x). \quad (18)$$

If C contains a global minimizer of (1), then F_C^* is the global optimal value.

Theorem 5.1 (basic consequences of SG-ReSC_C). *Assume (3), (4), and (9). Suppose that each iteration satisfies (6), that every shadow point lies in C , and that SG-ReSC_C holds. Then the following statements hold.*

(i) *For every $x \in C$ and every $k \geq 0$,*

$$\sum_{i=0}^k D_\phi(\tilde{x}^{i+1}, x^i) \leq \frac{\lambda D_\phi(x, x^0) + (k+1)(F(x) - F_C^*)}{\mu}. \quad (19)$$

(ii) *For every $x \in C$ and every $k \geq 0$, the averaged shadow iterate*

$$\bar{x}^{k+1} := \frac{1}{k+1} \sum_{i=0}^k \tilde{x}^{i+1}$$

satisfies

$$F(\bar{x}^{k+1}) - F(x) \leq \frac{\lambda}{k+1} D_\phi(x, x^0). \quad (20)$$

In particular, if $x^ \in S_C$, then*

$$F(\bar{x}^{k+1}) - F_C^* \leq \frac{\lambda}{k+1} D_\phi(x^*, x^0). \quad (21)$$

(iii) If $S_C \neq \emptyset$ and $x^* \in S_C$, then

$$\sum_{i=0}^{\infty} (F(\tilde{x}^{i+1}) - F_C^*) + \mu \sum_{i=0}^{\infty} D_{\phi}(\tilde{x}^{i+1}, x^i) \leq \lambda D_{\phi}(x^*, x^0). \quad (22)$$

Consequently,

$$F(\tilde{x}^k) \rightarrow F_C^*, \quad D_{\phi}(\tilde{x}^{k+1}, x^k) \rightarrow 0, \quad (23)$$

and SG-ReSC_C further implies

$$D_{\phi}(\tilde{x}^{k+1}, x^{k+1}) \rightarrow 0, \quad \delta_k + \omega_C(\Delta^k; \tilde{x}^{k+1}) \rightarrow 0. \quad (24)$$

Proof. Fix $x \in C$. From (14),

$$\mu D_{\phi}(\tilde{x}^{i+1}, x^i) \leq \lambda D_{\phi}(x, x^i) - \lambda D_{\phi}(x, x^{i+1}) + F(x) - F(\tilde{x}^{i+1}).$$

Since $\tilde{x}^{i+1} \in C$, we have $F(\tilde{x}^{i+1}) \geq F_C^*$. Summing from $i = 0$ to k gives (19).

Summing (14) again, now dropping the nonpositive term $-\mu D_{\phi}(\tilde{x}^{i+1}, x^i)$, gives

$$\sum_{i=0}^k F(\tilde{x}^{i+1}) - (k+1)F(x) \leq \lambda D_{\phi}(x, x^0) - \lambda D_{\phi}(x, x^{k+1}) \leq \lambda D_{\phi}(x, x^0).$$

Because C and F are convex, Jensen's inequality gives

$$F(\bar{x}^{k+1}) \leq \frac{1}{k+1} \sum_{i=0}^k F(\tilde{x}^{i+1}),$$

which proves (20) and (21).

Finally set $x = x^* \in S_C$ in (14). Then

$$F(\tilde{x}^{i+1}) - F_C^* + \mu D_{\phi}(\tilde{x}^{i+1}, x^i) \leq \lambda D_{\phi}(x^*, x^i) - \lambda D_{\phi}(x^*, x^{i+1}).$$

Summing over i proves (22). The limits in (23) follow because the summands in (22) are nonnegative. The limits in (24) follow from SG-ReSC_C, since each term on the left of (10) is nonnegative and the right-hand side tends to zero. $\square$

6 A last-shadow-iterate bound

The preceding theorem already gives convergence of the shadow objective values. A last-shadow-iterate rate can be recovered by the same weighted telescoping argument used in the zero-residual relative-error analysis.

Theorem 6.1 (last-shadow-iterate estimate). *Assume the hypotheses of Theorem 5.1. Assume in addition that $0 < \sigma < (\lambda - L)/\lambda$, that $\tilde{x}^0 = x^0 \in C$, and that every $\tilde{x}^k$ lies in C . Define*

$$b := \frac{\sigma\lambda}{\lambda - L - \lambda\sigma}, \quad a := (1+b)\lambda, \quad c := 2\lambda - \frac{\lambda - L}{\sigma}. \quad (25)$$

Then, for every $x \in C$ and every $k \geq 0$,

$$F(\tilde{x}^{k+1}) - F(x) \leq \frac{1}{k+1} \left[a D_{\phi}(x, x^0) + b(k+1)(F(x) - F_C^*) + c \sum_{i=0}^k i D_{\phi}(\tilde{x}^{i+1}, x^{i+1}) \right]. \quad (26)$$

If $S_C \neq \emptyset$, $x^* \in S_C$, and either

$$0 < \sigma \leq \frac{\lambda - L}{2\lambda} \quad (27)$$

or

$$\sum_{i=0}^{\infty} iD_{\phi}(\tilde{x}^{i+1}, x^{i+1}) < \infty, \quad (28)$$

then

$$F(\tilde{x}^k) - F_C^* = O(1/k). \quad (29)$$

Proof. SG-ReSC_C implies

$$D_{\phi}(\tilde{x}^{i+1}, x^{i+1}) \leq \sigma D_{\phi}(\tilde{x}^{i+1}, x^i). \quad (30)$$

Set $x = \tilde{x}^i$ in (14). This is legitimate because $\tilde{x}^i \in C$. Dropping the nonpositive term $-\lambda D_{\phi}(\tilde{x}^i, x^{i+1})$ and using (30), we get

$$\begin{aligned} F(\tilde{x}^{i+1}) - F(\tilde{x}^i) &\leq \lambda D_{\phi}(\tilde{x}^i, x^i) - \mu D_{\phi}(\tilde{x}^{i+1}, x^i) \\ &\leq \lambda D_{\phi}(\tilde{x}^i, x^i) - \frac{\mu}{\sigma} D_{\phi}(\tilde{x}^{i+1}, x^{i+1}) \\ &= \lambda D_{\phi}(\tilde{x}^i, x^i) - \lambda D_{\phi}(\tilde{x}^{i+1}, x^{i+1}) + c D_{\phi}(\tilde{x}^{i+1}, x^{i+1}), \end{aligned}$$

where the identity $c = \lambda - \mu/\sigma$ follows from (25). Set

$$G_i := F(\tilde{x}^i) + \lambda D_{\phi}(\tilde{x}^i, x^i), \quad i \geq 0.$$

The preceding inequality says

$$G_{i+1} \leq G_i + c D_{\phi}(\tilde{x}^{i+1}, x^{i+1}). \quad (31)$$

For every $i \geq 0$, (31) implies

$$G_{i+1} \geq (i+1)G_{i+1} - iG_i - ciD_{\phi}(\tilde{x}^{i+1}, x^{i+1}).$$

Summing from $i = 0$ to k and telescoping gives

$$\sum_{i=0}^k G_{i+1} \geq (k+1)G_{k+1} - c \sum_{i=0}^k iD_{\phi}(\tilde{x}^{i+1}, x^{i+1}) \geq (k+1)F(\tilde{x}^{k+1}) - c \sum_{i=0}^k iD_{\phi}(\tilde{x}^{i+1}, x^{i+1}). \quad (32)$$

On the other hand, the proof of Theorem 5.1 gives

$$\sum_{i=0}^k F(\tilde{x}^{i+1}) \leq (k+1)F(x) + \lambda D_{\phi}(x, x^0). \quad (33)$$

Moreover, by (30) and (19),

$$\begin{aligned} \sum_{i=0}^k D_{\phi}(\tilde{x}^{i+1}, x^{i+1}) &\leq \sigma \sum_{i=0}^k D_{\phi}(\tilde{x}^{i+1}, x^i) \\ &\leq b \left[D_{\phi}(x, x^0) + \lambda^{-1}(k+1)(F(x) - F_C^*) \right]. \end{aligned} \quad (34)$$

Combining (32), (33), and (34) yields (26).

If $x = x^* \in S_C$, then the term involving $F(x) - F_C^*$ vanishes. Under (27), one has $c \leq 0$, and the final term in (26) is nonpositive; hence (29) follows. Under (28), the final term is bounded independently of k , so (29) again follows. $\square$

7 Whole-sequence convergence

To obtain convergence of the iterates rather than only convergence of objective values, one needs the standard consistency assumptions connecting the Bregman distance and the ambient topology. The following formulation is deliberately stated as an assumption; it is satisfied by the usual Legendre kernels under the corresponding hypotheses used in Bregman proximal-gradient analyses.

Definition 7.1 (Bregman consistency assumptions). *We say that ϕ satisfies the consistency assumptions if:*

$$(B1) \quad \text{dom } \phi = \text{cl}(U).$$

$$(B2) \quad \text{For every } x \in \text{dom } \phi \text{ and every } \alpha \in \mathbb{R}, \text{ the set}$$

$$\{y \in U : D_\phi(x, y) \leq \alpha\}$$

is bounded.

$$(B3) \quad \text{If } y^k \in U \text{ and } y^k \rightarrow y \in \text{dom } \phi, \text{ then } D_\phi(y, y^k) \rightarrow 0.$$

$$(B4) \quad \text{If } u^k \in \text{dom } \phi \text{ is bounded, } y^k \in U, y^k \rightarrow y, \text{ and } D_\phi(u^k, y^k) \rightarrow 0, \text{ then } u^k \rightarrow y.$$

Theorem 7.2 (whole-sequence convergence). *Assume the hypotheses of [Theorem 5.1](#). Assume also the Bregman consistency assumptions in [Theorem 7.1](#), $S_C \neq \emptyset$, and boundedness of the shadow sequence $\{\tilde{x}^k\}$. Then there exists $x^\infty \in S_C$ such that*

$$x^k \rightarrow x^\infty, \quad \tilde{x}^k \rightarrow x^\infty. \quad (35)$$

Proof. Let $x^* \in S_C$. Setting $x = x^*$ in [\(14\)](#) gives

$$D_\phi(x^*, x^{k+1}) \leq D_\phi(x^*, x^k) - \lambda^{-1}(F(\tilde{x}^{k+1}) - F_C^*) - \lambda^{-1}\mu D_\phi(\tilde{x}^{k+1}, x^k). \quad (36)$$

Thus $\{D_\phi(x^*, x^k)\}$ is nonincreasing and bounded below. By (B2), $\{x^k\}$ is bounded. Hence $\{x^k\}$ has a cluster point; choose a subsequence $x^{k_j} \rightarrow z$.

By [Theorem 5.1](#),

$$D_\phi(\tilde{x}^{k+1}, x^k) \rightarrow 0, \quad D_\phi(\tilde{x}^{k+1}, x^{k+1}) \rightarrow 0, \quad F(\tilde{x}^k) \rightarrow F_C^*. \quad (37)$$

The boundedness of $\{\tilde{x}^k\}$, the convergence $x^{k_j} \rightarrow z$, and (B4) applied to $u^j = \tilde{x}^{k_j+1}$ and $y^j = x^{k_j}$ imply

$$\tilde{x}^{k_j+1} \rightarrow z.$$

Since C is closed and all shadow iterates lie in C , we have $z \in C$. Lower semicontinuity of F gives

$$F(z) \leq \liminf_{j \rightarrow \infty} F(\tilde{x}^{k_j+1}) = F_C^*.$$

Because $z \in C$, the reverse inequality $F(z) \geq F_C^*$ is automatic; hence $z \in S_C$.

Now repeat the Fejer inequality [\(36\)](#) with x^* replaced by this optimal cluster point z . It follows that $\{D_\phi(z, x^k)\}$ is convergent. Along the subsequence $x^{k_j} \rightarrow z$, assumption (B3) gives $D_\phi(z, x^{k_j}) \rightarrow 0$. Therefore the full sequence satisfies

$$D_\phi(z, x^k) \rightarrow 0. \quad (38)$$

If w is any cluster point of $\{x^k\}$, say $x^{\ell_j} \rightarrow w$, then (B4), applied to the constant sequence $u^j = z$ and $y^j = x^{\ell_j}$ together with [\(38\)](#), implies $w = z$. Thus $x^k \rightarrow z$.

Finally, $x^k \rightarrow z$, $D_\phi(\tilde{x}^{k+1}, x^k) \rightarrow 0$, boundedness of $\{\tilde{x}^k\}$, and (B4) imply $\tilde{x}^{k+1} \rightarrow z$. Hence [\(35\)](#) holds with $x^\infty = z$. $\square$

8 Conclusion

The zero-residual assumption in the non-inertial partial relative stopping criterion can be weakened in a precise comparison-set sense. The proof does not require a new descent argument; it requires keeping the residual term with its sign and absorbing its worst-case value over C into the relative Bregman error budget. The resulting criterion gives the same telescoping estimates as the original zero-residual theory.

This is genuine partial progress on the question raised by Yang and Toh, but it is not a complete solution. It gives a nonzero-residual relative criterion for the non-inertial method in bounded or otherwise support-controlled comparison regions. It does not solve the unrestricted case $C = \mathcal{E}$ with arbitrary nonzero residuals, and it does not cover the inertial variant.

References

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